



Gerry goes to sleep at 11 pm each night. But he doesn't sleep very well, so he awakens at a random time between 1:00 am and 3:59 am, with each minute equally likely (including 1:00 and 3:59). He then reads the hour and minute shown on his clock as a three digit number, so 2:56 am would be 256. What is the probability that that number will be divisible by 7?"

Debate Round 1

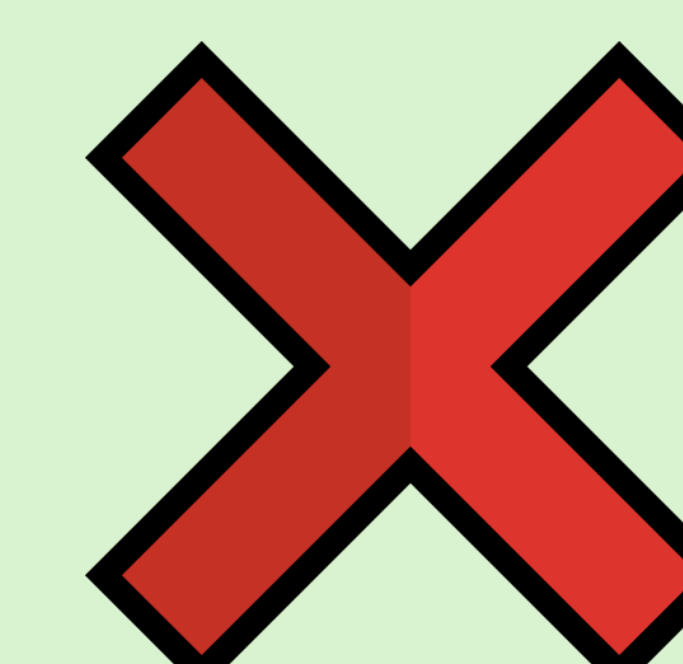
Agent 1

- Gerry can wake up anytime between 1:00 and 3:59 → 180 possible minutes.
- Each minute is written as a three-digit number (e.g., 1:00 → 100).
- Count how many of these are divisible by 7:
 - 1:00–1:59 → 8 numbers,
 - 2:00–2:59 → 9 numbers,
 - 3:00–3:59 → 9 numbers.
- Total favorable = 26.
- Probability = $26/180 = 13/90$.



Agent 2

- Gerry can wake up anytime from 1:00 to 3:59 → 180 possible minutes.
- Each time is written as a three-digit number (HMM).
- Find numbers divisible by 7 between 100 and 359.
- Smallest is 105, largest is 357 → total 37 valid times.
- Probability = $37/180$.

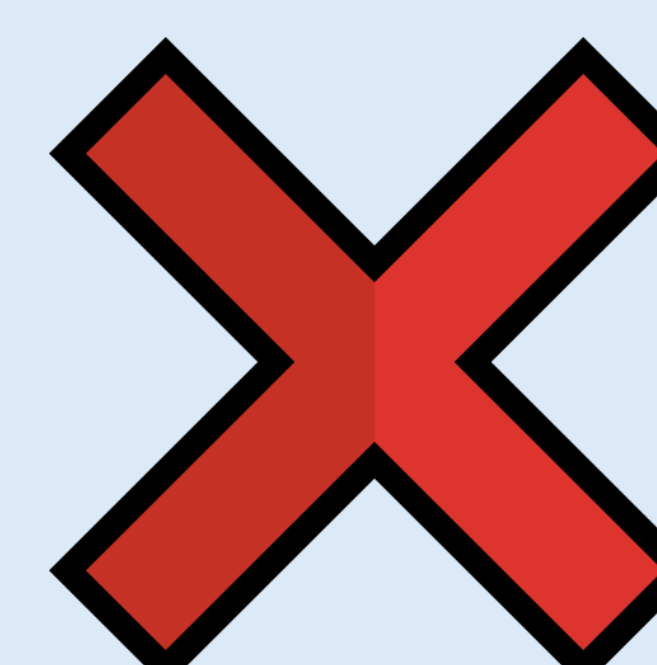


Take both responses as references

Debate Round 2

Agent 1

- Gerry can wake between 1:00 and 3:59 → 180 possible minutes.
- Each time is written as a three-digit number (100–359).
- Numbers divisible by 7 in this range go from 105 to 357 → 37 valid cases.
- Probability = $37/180$.



Agent 2

- Gerry can wake between 1:00–3:59 → 180 total minutes (180 possible numbers from 100–359).
- Numbers divisible by 7 in this range are from 105 to 357.
- There are 37 such valid times.
- Probability = $37/180$.

